



Transportation Security Administration
Request for Information

Alarm Resolution and On-Person Screening Portfolios

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Submission Deadline

Event	Date	Submission Requirements
Response to Request for Information (RFI)	May 22, 2025, 3PM Eastern Daylight Time (EDT)	Refer to Section 4.0, Submission Details



REQUEST FOR INFORMATION

1.0 General Information

1.1 Subject

Request for Information (RFI) for the Department of Homeland Security (DHS) Transportation Security Administration (TSA) Alarm Resolution (AR) and On-Person Screening (OPS) Portfolios regarding the following Transportation Security Equipment (TSE):

- Bottled Liquid Scanner (BLS)
- Advanced Imaging Technology (AIT)
- Explosives Trace Detection (ETD)

1.2 Introduction

The AR portfolio, which oversees secondary screening equipment, and the OPS portfolio, which oversees people screening equipment, within the TSA's Acquisition Program Management (APM) Office, are requesting information and conducting market research on potential sources for BLS, AIT and ETD technologies and associated support services.

The TSA is issuing this RFI as part of its market research efforts to determine business interest and capability of potential sources for the manufacture, servicing, and support of BLS, AIT, and ETD systems. Servicing and support include maintenance, configuration management, and any system specific test items, among other services.

This RFI is issued solely for market research, information, and planning purposes; it does not constitute a Request for Proposal (RFP) or any commitment or intent to issue an RFP. The RFI does not commit the TSA to a contract for any supply or service. The TSA is not currently seeking proposals. Responders are advised that the TSA will not pay any costs incurred in response to this RFI. All costs associated with responding to the RFI will be solely at the interested party's expense. Failure to respond to this RFI does not preclude participation in any future RFP.

The information provided in this RFI is subject to change and is not binding on the TSA. All submissions become the property of the TSA and will not be returned. TSA is requesting information to identify BLS, AIT, and ETD solutions that are currently available or under development.

The TSA is interested in information from all parties, including manufacturers as well as those researching system components and subsystems.

1.3 Agency Name

Department of Homeland Security
Transportation Security Administration



6595 Springfield Center Dr., Springfield, VA 20598

1.4 Program Information

RFI Number: TSA25-04-0463

Program Names: Bottled Liquid Scanner, Advanced Imaging Technology, and Explosives Trace Detection

Managing Portfolio: Alarm Resolution and On-Person Screening

2.0 Background

The mission of the TSA is to protect the nation's transportation systems to ensure freedom of movement for people and commerce. To achieve its mission, the TSA must ensure that its security screening equipment is both effective and efficient. This includes security equipment used at all domestic transportation facilities, including aviation, rail, other surface transportation, and maritime transportation (including port security). The TSA's AR portfolio is responsible for secondary screening of potential threats, and the OPS portfolio is responsible for screening people for potential threats. TSA is continuously evaluating and modernizing processes and capabilities to strengthen the agency's ability to respond to adversarial actors, detect emerging and evolving threats, and enable efficient and effective targeting of screening resources.

3.0 Objectives for TSE

In a diverse and rapidly evolving threat environment, TSA's security effectiveness strategies must achieve maximum future-state efficiency. To enable this mission objective, TSA is looking to establish enhanced operational and detection capabilities, which are outlined below.

3.1 Bottled Liquid Scanner

The BLS is equipment designed to detect a wide range of hazardous and prohibited substances in liquids, aerosols, and gels contained in bottles or other sealed containers. Any liquid, aerosol, gel, cream, or paste that alarms during the initial screening process requires secondary screening. The BLS can identify hazardous substances, including explosives and toxic chemicals using advanced detection technologies such as Raman spectroscopy and infrared analysis. The BLS can differentiate between safe liquids like water, juice, or shampoo and dangerous substances without opening the container, ensuring minimal disruption at security checkpoints. It can analyze a variety of container materials, including plastic, glass, and metals, and can effectively scan liquids of varying viscosities and compositions. Additionally, the BLS provides rapid and accurate results, making it suitable for high throughput environments such as the security checkpoint.

TSA is looking for BLS solutions for the screening of closed containers while optimizing operational efficiency and throughput rate. The BLS RFI is focused on gathering responses for the following objectives:

- Detection Capabilities
 - Enhanced Threat Detection
 - Material Composition Analysis
 - Low Probability of False Alarms



- Adaptability to emerging threats
- Container Compatibility
 - Capable of analyzing containers of various materials
 - Handles containers up to 38 cm (15 inches) high and 20 cm (8 inches) wide
 - Liquid content range of 100ml-2000ml
- Efficiency
 - Fast, automated screening
 - Low maintenance
 - User interface friendly
 - Easy training
- Safety
 - Non-ionizing fields
 - Operator-safe

3.2 Advanced Imaging Technology

TSA is looking to establish OPS solutions with a focus on enhancing threat detection capabilities with reduced false alarm rates to reduce contact between passengers and TSOs, moving passengers through the checkpoint in a continuous manner, standardizing image display, and enabling operational connectivity through secure data transmission. The OPS RFI is focused on gathering responses for the following objectives:

- Enhanced detection performance to meet new detection standards
 - Identify potential threats and prohibited items regardless of material composition, including metallic and non-metallic objects.
- Material discrimination

TSA is looking to develop or integrate capabilities and technologies to enhance material discrimination of OPS systems.

 - Methods to enhance detection performance may include, but are not limited to:
 - Developing/improving machine learning algorithms
 - Screening options that utilize non-ionizing portion of the EM spectrum or non-EM technologies
 - Desired threshold and objective end-states are provided below:
 - Systems that achieve threshold performance will have the ability to detect the full list of threats, per detection standards, and be able to classify alarms as metallic or non-metallic objects.
 - Systems that achieve objective performance will have the ability to identify additional material categories for alarms, including but not limited to water, ceramics, plastics, paper, leather, carbon fiber, and potential explosive material in accordance with the latest detection standard.
- Footwear screening to eliminate the need to remove shoes
 - TSA is looking for AIT equipment capable of rapidly screening footwear without requiring footwear removal or modification by the person. The speed of screening, which should not slow individuals down, and final unit cost are major factors for the adoption of this type of OPS technology. Foot anatomical details



may not be shown to a TSO, so an automated threat detection algorithm must be used.

- The technology must fit in the current screening lane space and be integrated with or placed in front of the current AIT systems. It is understood that certain types of footwear may be more difficult to screen than others as a result of the materials used in their construction. At a minimum, 80% of the footwear of the traveling public must be capable of being screened and meeting requirements listed in the latest applicable detection standard. The type of footwear that is not able to be screened must be identified by the vendor.
- Desired threshold and objective end-states are provided below.

Systems that meet threshold performance shall meet at least one of the threshold criteria in Table 3.

Table 3: Threshold Performance Criteria for Footwear Screening Criteria

	Detection Standard	Throughput (Minimum Average)	Additional Criteria
Threshold 1	Level 2 equivalent	200 passengers/hour	-
Threshold 2	Level 2 equivalent	360 passengers/hour	

Systems that meet objective performance shall meet at least one of the objective criteria in Table 4:

Table 4: Objective Performance Criteria for Footwear Screening Criteria

	Detection Standard	Throughput (Minimum Average)	Additional Criteria
Objective 1	Level 4 equivalent	200 passengers/hour	
Objective 2	Level 3 equivalent	360 passengers/hour	

- Enhanced throughput to facilitate increasing passenger volume
 - Implementing automated/high throughput design by scanning as individuals pass through while fully clothed, without the need to remove belongings, outer garments, or the need to stop walking to assume specific postures.
 - Utilizes AI-based algorithms to automatically detect threats, displaying results on a sex-neutral avatar to ensure privacy.
 - Supports TSOs touchless resolution capability and via option to fully view system identified threats via two more three-dimensional representative human figures.
- Improved data visualization

TSA wants to enhance TSO ability to process individuals by improving how data is displayed to them. The process for TSOs to review security scanning data is complex. Improved data visualization tools may have a large impact on a TSO's image interpretation performance, error rates, efficiency, and cognitive load. Two methods of interest for improved data visualization are a Common Graphical User Interface (CGUI) and alternative data visualization methods.



- Improved data visualization should generate high-resolution, two or three-dimensional representative human figures in real-time, enabling immediate analysis and response. Novel or emerging data visualization capabilities have the potential to enhance TSO operations by providing additional sensor capability and connectivity with multiple checkpoint systems, improved surveillance functions, and a means to improve Standard Operating Procedure (SOP) compliance. Current OPS technologies rely heavily on the TSO for material analysis and identification of abnormalities. Data visualization tools can support TSOs by enhancing visual inspection. An improved surveillance function could augment security by turning operators into a network of relayed information via analytical sensors and archived video feed to allow for “person to item” tracking that ensures accurate baggage custody (i.e., visual overlays of threats on closed circuit television feeds). Lastly, improved SOP compliance can be achieved by providing operators instructions in real time for unique scenarios faced at the checkpoint.
 - Desired threshold and objective end-states are provided below.

Systems that meet threshold performance shall meet at least one of the threshold criteria in Table 1.

Table 1: Threshold Performance Criteria for Enhanced Detection & Throughput Criteria

Criteria	Detection Standard	Throughput (Minimum Average)	Additional Criteria
Threshold 1	Level 2	200 passengers/hour	
Threshold 2	Level 2	200 passengers/hour	- No need to divest light outerwear
Threshold 3	Level 2	360 passengers/hour	

Systems that meet objective performance shall meet at least one of the objective criteria in Table 2.

Table 2: Objective Performance Criteria for Enhanced Detection & Throughput

	Detection Standard	Throughput (Minimum Average)	Additional Criteria
Objective 1	Level 4	200 passengers/hour	
Objective 2	Level 3	200 passengers/hour	- No need to divest light outerwear
Objective 3	Level 3	360 passengers/hour	

- Safety

TSA wants AIT devices to operate using non-ionizing radio waves, enabling non-contact passenger screening and making it safe for all individuals, including those with wheelchair, personal medical device(s), and health concern(s). Desired features include:

 - Incorporation of International Electrotechnical Commission (IEC) standard 63391 Active Millimeter- Wave Systems for Security Screening of Humans – General Requirements



- New countermeasure systems and processes
TSA is also looking for solution developers and vendors to propose novel countermeasure or checkpoint systems, processes, and/or designs. Checkpoints vary with airport size and use different lane types and configurations to screen passengers.
- Synthetic data creation and the methods used to achieve it (e.g., artificial intelligence, ray tracing techniques, computational electromagnetic methods)
TSA is exploring concepts and approaches to quickly create synthetic data to supplement real data for training and testing of machine learning algorithms for OPS systems. The intent is that this tool would effectively modify existing scans of individuals with and without threats on their bodies. The system must be easily adapted for different system types, such as EM frequencies, antenna configurations, radiation patterns, and beam widths. The vendor must show evidence that machine learning algorithms would reasonably not be able to tell the difference between the synthetic and real data.

Desired threshold and objective end-states are provided below:

Systems that achieve threshold performance will result in a tool capable of removing and placing new threats of varying shapes and sizes on 3D and 2D scans of individuals of varying body shapes and sizes. The system must be capable of changing the threat's location on the body and orientation, mimicking realistic concealment scenarios.

- Incorporation of Institute of Electrical and Electronics Engineers (IEEE) N42.59 image quality standard with characterization of metric thresholds causing performance degradation.
- Ability to realize intersection over union as a target indication to the operator and the accuracy and precision of the capability.
- System architecture flexibility
TSA wants the system architecture to be integrated into its existing security infrastructure, providing scalability for different locations and needs. Such systems must:
 - Align to ongoing plans for integrating all screening systems with checkpoint TSE
 - Maintain robust cybersecurity posture
 - Promote connectivity, modularity, and open architecture to support enhancements (e.g., 3rd party algorithms, common interfaces, standardized data formats, etc.)
 - Improve security capabilities by increasing the speed of acquiring or enhancing TSE and reducing the difficulty for vendors to meet TSA requirements by establishing common interfaces between TSE subcomponents.
 - Promote connectivity, modularity, and open architecture (OA), which enables new industry-developed solutions by creating the tools to implement an open and modular system architecture. OA incorporates standardized data formats, such as Digital Imaging and Communications in Security (DICOS); standardized interfaces, such as the Software Development Kit (SDK) and application programming interfaces (APIs) in the Open Platform Software Library (OPSL)1;



and systems of systems support with workflow logic that allows different TSEs to be interconnected.

- Practice of National Electrical Manufacturers Association (NEMA) IIC 1 v05, Digital Imaging and Communications in Security (DICOS) standard and image curation methodology
- Adapt to application program interfaces use of software development kits, use of common Open Platform Software Libraries, graphical user interfaces and automated target recognition algorithms.
- Have the capacity and flexibility of an AIT's computer architecture to exist locally on the unit or on a central command and control station, such as a threat recognition server where multiple units of transportation security equipment shares a server resource.
- Usage of existing infrastructure at a checkpoint to reduce total cost of ownership.

3.3 Explosives Trace Detection

ETD technology is used to screen travelers and their property. ETD systems identify trace threats by detecting the chemical attributes/signatures of microscopic residue of threat compounds. The technology is highly sensitive, thereby enabling fast and accurate screening for trace explosives on a variety of surfaces and can be used in all public areas and airports to screen people and their accessible property for the presence of trace threats. The operational procedure includes the collection and analysis of a sample. The ETD system then informs the user both visually and audibly if the sample contains a prohibited threat(s). TSA's goal is to enable fast and accurate screening for trace threats on a variety of surfaces. The ETD RFI is focused on gathering responses for the following objectives:

- Enhanced Threat Detection Performance and Throughput
- Capability to Add Additional Threats to the Threat Library
- Low False Alarm Rate
- Enhanced Reliability, Maintainability and Availability (RMA) Metrics
 - Reliability is the measure that determines the frequency of system failures or maintenance actions and supports the user need for satisfactory performance for a given period.
 - Maintainability is the ability of the system to be restored to operational condition when corrective maintenance is performed by personnel having specified skill levels, using prescribed procedures and resources, at each prescribed level of maintenance and repair. It includes the total corrective maintenance time to repair the system, but it does not include lead time for parts not readily available or other administrative or logistics downtime.
 - Availability is the percentage of time, over a period during operational hours, that TSE is available to perform its required mission. It reflects all potential causes of downtime, such as administrative downtime, logistics downtime, etc., and total TSE operational uptime.
- Inclusion of a Field Data Reporting System (FDRS)
 - An FDRS refers to a software or platform that allows users to collect and submit data directly in real-time, capturing information about observations,



measurements, or events, and then presenting that data in a structured format for analysis and reporting back to a specific system location; essentially, it's a tool for streamlining the process of gathering data and transmitting it for further use.

- Operator Interface Easy to Use
- Audible and Visual Indicators
- Safety
 - Systems do not expose operators or passengers to risks or hazards such as electrical, temperature, ergonomic stressors, radioactivity, or lasers.

4.0 Responses to this RFI

Vendor response(s) may be releasable to the public under the Freedom of Information Act (FOIA), 5 U.S.C. 552, and the DHS FOIA regulation, 6 C.F.R. Chapter 1 and Part 5. Therefore, please segregate and mark proprietary information. Proprietary information should be minimized and **MUST BE CLEARLY MARKED**. Please be advised that all submissions become the property of the Federal Government and will not be returned.

The information submitted in support of this RFI must be submitted in writing to the points of contact identified below. Responses must include a cover letter with the following information:

- Company's Name
- Company's Address
- Unique Identifier Number (UEI) Number
- Company's Size and Socio-Economic Status information
- Technical Point of Contact(s) (Name, title, email, and phone number)

The following sections provide a recommended outline for a response to this RFI.

Section 1 – Corporate Expertise (1-3 pages)

Briefly describe the company, products and services, history, ownership, business size, and other information deemed relevant. Include any past projects supporting the TSA and a description of tasks performed.

Section 2 – Capabilities and Conceptual Alternatives (1-9 pages)

Although the TSA is not providing any Sensitive Security Information (SSI) as a result of this RFI, all responses to the below attachments **MUST** be marked as SSI per Code of Federal Regulations part 1520.13 ([see guidance here](#)). Please ensure all submissions to the below attachments are password protected, the password sent in a separate email, and all documents have the required SSI header and footer.

Attachment 1 – BLS RFI Questions provides a recommended outline for response to this RFI relating to BLS.

Attachment 2 – ETD RFI Questions provides a recommended outline for response to this RFI relating to ETD objectives.



Attachment 3 – AIT RFI Questions provides a recommended outline for response to this RFI relating to AIT objectives. TSA requests a written response that addresses each of the objectives noted in Attachment 1, Attachment 2, and/or Attachment 3 to better understand market capabilities. The questions in the attachments are intended to minimize the effort of the respondent and structure the responses for ease of analysis by the Government.

The vendor is not required to answer or comment on all content contained in the attachments in order to respond to this notice. Interested vendors are encouraged to respond to all or some of the topics presented for BLS, AIT, or ETD.

Adobe pdf attachments such as test reports and product brochures may be submitted as part of the response, but no more than 15 pages may be submitted in addition to the attachments. Electronic file sizes should not exceed 10 Megabytes (MB).

5.0 Disclaimer

This RFI is issued for information and planning purposes only and does not constitute a solicitation. All information received in response to this RFI that is marked Proprietary will be handled accordingly. The Government will not return or pay for any information provided in response to this announcement; no basis for a claim against the Government shall arise as a result from a response to this notice or Government use of any information provided. Responders are solely responsible for all expenses associated with responding to this announcement.

6.0 Review of Vendor Responses

Only electronic submissions via email will be accepted. Each electronic submission shall include the TSE of interest in the subject line followed by RFI Capabilities Response (e.g., “BLS RFI Capabilities Response”). In the body of the email, include a vendor Point of Contact (POC), phone number and email address of the person to be contacted regarding any correspondence between the Government and the vendor. This person should be a primary contact and capable of addressing questions or issues associated with the submission and content of the RFI Capabilities Responses.

7.0 Vendor Submission of Responses and Contact Information

Please follow the following response format:

- The TSA requests a written response to address any or all questions above. Responses should not exceed 12 pages in total. The written response shall be in either read-only Word or PDF format.
- Vendors may submit test reports and product brochures as Adobe PDF attachments, limited to 15 additional pages.
- Responses should not be greater than 27 pages in total, including any test reports or brochures.
- Electronic file sizes should not exceed 10 Megabytes (MB).
- Respondents should also include contact information should TSA have questions or want to hold further discussion.



- Written responses shall be submitted no later than **May 22, 2025, at 3:00pm EDT to the respective Program Management Office as follows:**
 - **BLS – William Jones at BLS@tsa.dhs.gov**
 - **AIT – Thi Nguyen at Thi.Nguyen@tsa.dhs.gov**
 - **ETD – Marilyn Gray at Marilyn.Gray@tsa.dhs.gov**

Important Note: The vendor is not required to answer or comment on all content contained in this RFI in order to respond to this notice. Interested vendors are encouraged to share ideas and suggestions to all or some of the topics presented herein.

The Government will review vendor responses for market research purposes only. The Government does not intend to provide a response to submissions for this RFI, but the Government reserves the right to hold a technical information exchange meeting after the RFI review at the agency's discretion. These meetings will be scheduled at the discretion of the Government and may not be scheduled with all vendors that respond to the RFI. In addition to Government review, the TSA reserves the right to have TSA support contractors provide advisory services to review responses to this RFI. Therefore, it will be necessary to allow the support contractors to have access to all responses received from this RFI. As such, vendors that intend to submit proprietary information to TSA must execute an agreement with each of the below firms that states that they will: (1) protect the vendor's information from unauthorized use or disclosure for as long as it remains proprietary, and (2) refrain from using the information for any purpose other than that for which it was furnished:

RELI Group
5520 Research Park Dr #105
Catonsville, MD 21228
RELI Phone: (866) 996-7354
Email Point of Contact: Margaret Jennings
Email: Margaret.Jennings@religroupinc.com

Global Systems Technologies, Inc.
109 Floral Vale Blvd
Yardley, PA 19067
GST Phone: (215) 579-8200
Email Point of Contact: Winifred DeSimone
Email: WiniD@gstpa.com

If proprietary information is to be provided to the TSA as part of the response to this RFI, agreements with the above contractor support companies must be provided as part of the response submission.



8.0 List of Acronyms

Acronym/Abbreviation	Definition
AIT	Advanced Imaging Technology
APM	Acquisition Program Management
AR	Alarm Resolution
ATO	Authority to Operate
BLS	Bottled Liquid Scanner
C&A	Certification and Accreditation
DHS	Department of Homeland Security
ETD	Explosives Trace Detection
FDRS	Field Data Reporting System
FSS	Federal Supply Schedule
GSA	General Services Administration
MB	Megabytes
NDA	Non-Disclosure Agreement
POC	Point of Contact
OPS	On-Person Screening
RMA	Reliability, Maintainability and Availability
RFI	Request for Information
RFP	Request for Proposal
TSA	Transportation Security Administration
TSE	Transportation Security Equipment
UEI	Unique Identifier Number